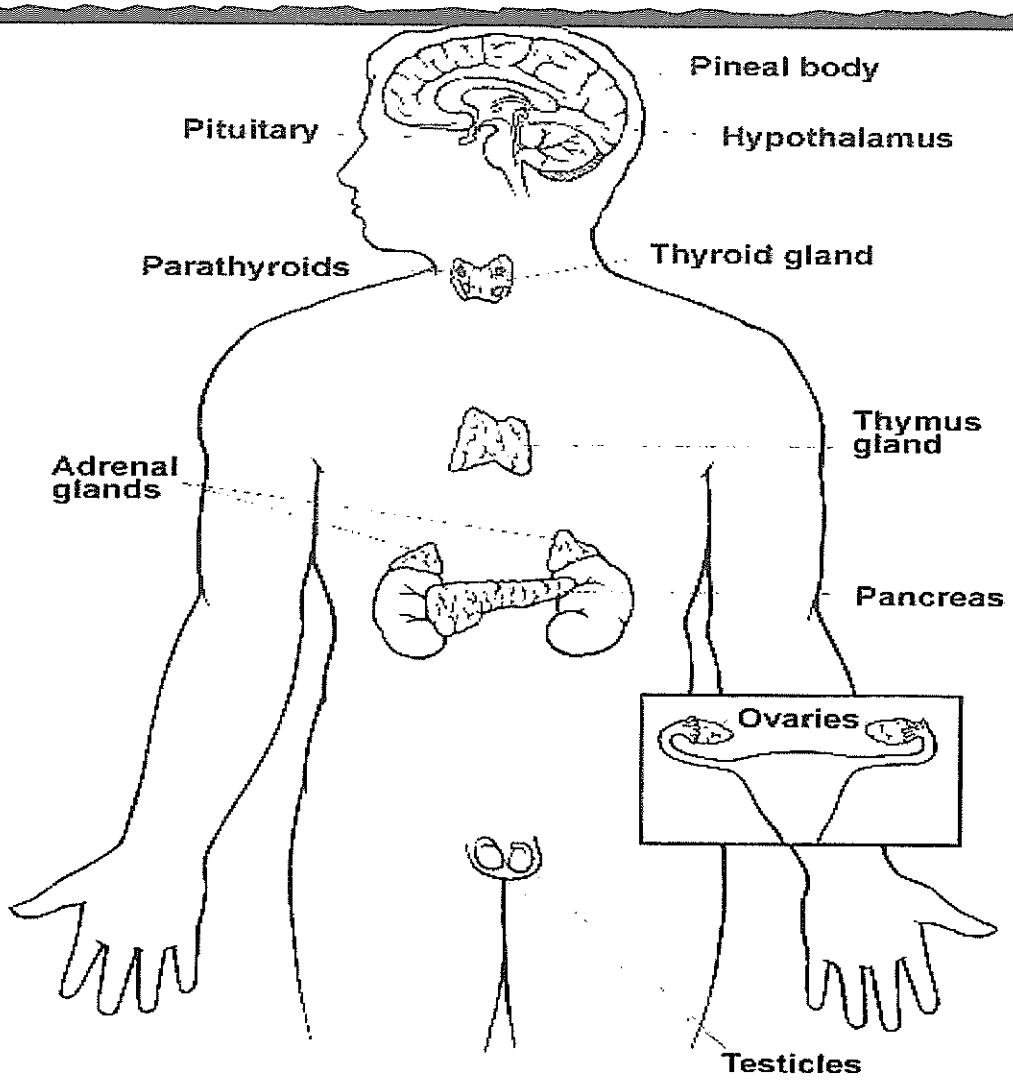
I'm getting Hormonal!!!!!

Hormones are chemical messengers, which help in controlling and coordinating the activities of the body. They are produced by endocrine glands.
The human body contains 30 hormones, which regulates activities like sleep, body temperature, hunger, and managing stress in times of crisis and so on.

The endocrine system of human beings consists of the following glands:

1. Hypothalamus and Pituitary Gland
2. The Thyroid Gland
3. The Parathyroid Glands
4. The Adrenal Glands
5. Islets Of Langerhans (Pancreas)
6. Ovaries (Female)
7. Testes (Male)


Pineal body
Pituitary
Hypothalamus
Parathyroids
Thyroid gland
Thymus gland
Adrenal glands
Pancreas
Ovaries
Testicles

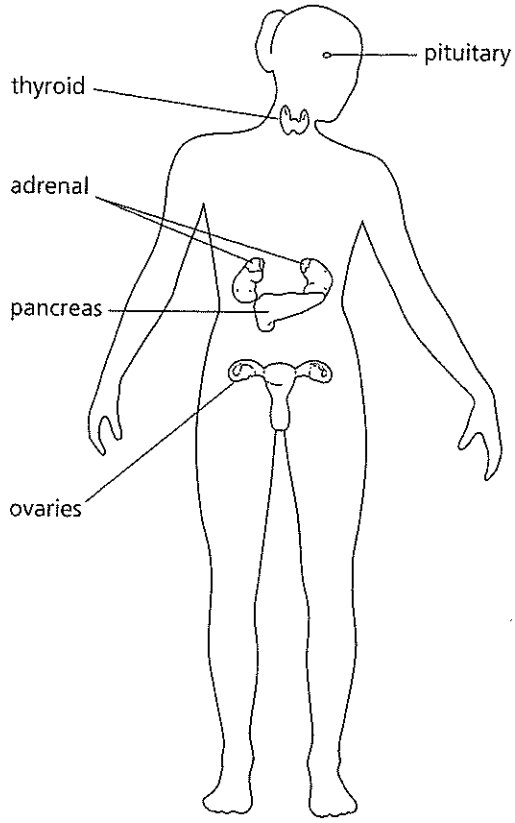
Hormones in the body

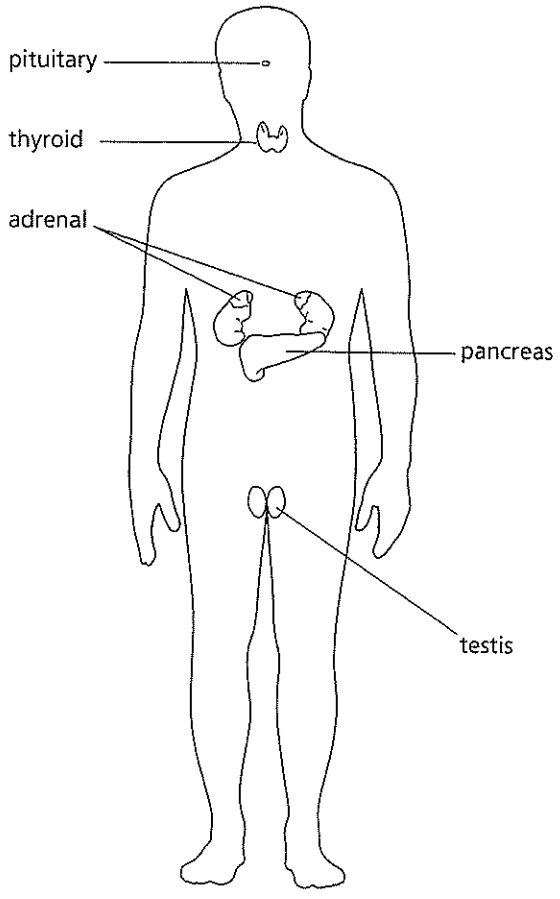
9a

Refer to the labelled diagrams below offering information about hormones and glands in the human male and female body.

Female

Pancreas Special cells found in the pancreas, called Langerhans, produce the hormone called insulin.
Insulin lowers the level of glucose in the blood, causing the liver and muscle cells to convert glucose to glycogen.
Pituitary gland is a very small gland, about the size of a pea. It produces human growth hormone which regulates growth in children.
Thyroid gland produces a hormone called thyroxine, which controls energy release in tissues.
Ovaries produce the sex hormones called oestrogen and progesterone which work the menstrual cycle and also develop the female body.
Adrenal gland produces adrenaline, which prepares the body for action by targeting the muscles, heart, liver and blood vessels.





Male

Male

Pituitary gland
- found at base of brain
- master gland
- controls all other glands
- lack of growth hormone could stunt growth permanently

Pancreas
- produces insulin
- not enough or too much insulin results in diabetes

Thyroid gland
- located in neck
- produces thryoxine
- controls metabolism (the rate at which tissues use energy)
- lack of iodine can result in a disease called goiter (enlarged thryroid)

Testis
- produces sex hormone, testosterone
- controls sexual function throughout the male body
- develops male body characteristics

Adrenal gland
- found near top of each kidney
- adrenalin is produced when stressed or excited
- adrenalin causes increase in heart beat, breathing, blood pressure and sugar levels in the blood

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Who's who in the hormone world??

GLAND	HORMONE SECRETED	TARGET ORGAN	FUNCTION
ANTERIOR PITUITARY	Growth Hormone (GH)	Ends of long bones	Growth.
ANTERIOR PITUITARY	Thyroid Stimulating Hormone (TSH)		
ANTERIOR PITUITARY	Follicle Stimulating Hormone (FSH)		
ANTERIOR PITUITARY	Luteinising Hormone (LH)		
ANTERIOR PITUITARY	Prolactin		
POSTERIOR PITUITARY	Oxytocin		
POSTERIOR PITUITARY	Anti-Diuretic Hormone (ADH)		
THYROID	Thyroxine		
PANCREAS Beta Cells	Insulin		
PANCREAS Alpha Cells	Glucagon		
TESTES	Testosterone		
OVARY	Oestrogen Progesterone		

It takes two to coordinate 10a

1 Below are four paragraphs of text about the two coordination systems in humans. Order the paragraphs by writing numbers in the *Order* column below.

Order	Paragraphs
	Both systems carry information from one place to another. Nerves are very quick. Neurones carry electrical impulses along their axons which then send transmitter chemicals from one axon to another. The impulses travel very quickly along the long axons, but the transmitter chemicals are much slower as they travel across the tiny gap between the dendrites. On the other hand the endocrine system places chemicals called hormones into the blood stream, which then takes them around the whole body at a fairly low speed. Several organs respond to the hormones, while others do not. Those target organs respond fairly slowly and the response can last for a long time, anything from a few minutes to a few months. However, the nervous impulse results in an immediate response which is short lived. The nerves take a message directly to a particular organ or tissue which alone responds.
	Here are some examples of responses involving one system or the other: When the light receptors in your eye relay a close and fast moving insect to the brain via the sensory neurones, the connector neurons in the central nervous system send a message along the motor neurones to the eye lid muscles. As a result of this reflex the lid closes, thus protecting the eye. When something hot or sharp touches you, you very quickly pull back that touched part of the body. Only nerves are involved in this quick and direct response. However, when the thyroid gland in your neck releases its hormones into the blood stream, many parts of your body react over the next few weeks and you grow taller and/or bigger. Another hormone is insulin. When insulin is released from the pancreas into the blood system, the cells around the body absorb glucose sugar from the blood. This glucose is then used for all kinds of cellular processes including respiration.
	In humans the nervous system and the endocrine system work together in order to control the body. Sometimes the nerves bring about responses, other times the hormones. How are they different? How do they work together?
	The fright response involves both systems. This response can be initiated by shock, anger, passion, anxiety or stress. The nervous system detects fright stimuli, which are relayed to the hypothalamus inside the brain. This stimulates the pituitary gland to release hormones into the blood stream. In particular, the adrenal glands above the kidneys react and push the hormone adrenalin into the blood. As a result your pupils dilate, you become pale, your sense organs become highly sensitive, your heart beats faster, glucose is released into the blood, your blood pressure increases and your lungs work more effectively. Your body is now ready for action: either for flight or for fight! Many mammals also have this fright response.

2 Each of the four paragraphs above covers one or more topics. In the table below, reveal the structure of the text by linking the main topics and subtopics to the ordered paragraphs.

Paragraph	Main topic	Subtopics

3 a State whether the text above is a discussion, an explanation, a procedure, a recount or a report.

_____

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It takes two to coordinate

b Justify your answer. _____

_____

4 Make a summary table showing similarities and differences between the nervous and endocrine systems. Use the *Aspect* column to identify the area you are looking at, then fill in the specific features of the *Nervous system* and the *Endocrine system*.

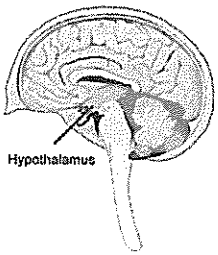
Aspect	Nervous system	Endocrine system
Purpose		
Speed		
Messengers		
Pathway		
Response organs		
Response activity		
Examples		

5 Discuss whether the nervous system or the endocrine system is more important to the human body. Use the scaffold below to help you. Remember to use scientific understanding to your advantage and to include exaggerated and emotional terms.

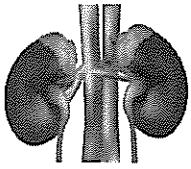
Title	
Issue	
Arguments for nervous system	
Arguments for endocrine system	
Conclusion	

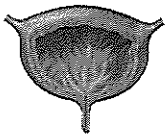
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Low Water Levels



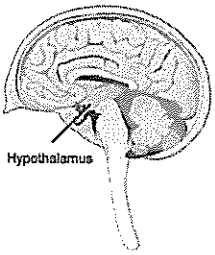


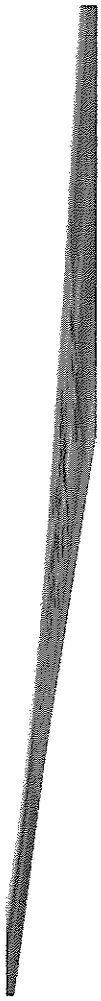


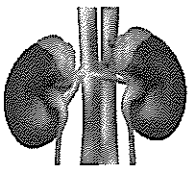


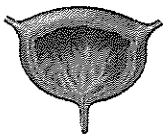


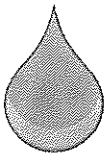
High Water Levels











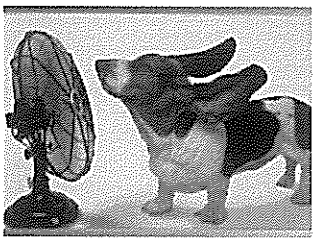
When you are Feeling Cold



What does the Endocrine System do?

What does the Nervous System do?

When you are Feeling Hot



What does the Endocrine System do?

What does the Nervous System do?